PUV Pipeline — Methods and Validation Results

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1 Methods

1.1 Nearshore wave measurements

Nearshore wave and current dynamics were measured using bottom-mounted Nortek Vector acoustic Doppler velocimeters (ADVs) configured in pressure–velocity (PUV) mode, sampling continuously at 2 Hz. Instruments were deployed along MOP (Monitoring and Prediction) transect lines at nominal depths of 5–7 m, with two cross-shore positions per transect. Deployment metadata—including instrument heading, clock drift, and pressure sensor height above the bed (z_s, typically 0.75–0.77 m)—were obtained from field deployment notes.

1.2 Level 1: Quality control and coordinate rotation

Raw burst files (.dat/.sen/.hdr) were merged into continuous time series at the 2 Hz sampling rate. Clock drift corrections were applied linearly over each deployment based on measured drift at recovery. A battery cutoff detection algorithm identified intermittent sampling caused by battery depletion by flagging gaps exceeding 1 s between consecutive sensor records; all data following the first such gap were discarded.

Quality control filters removed samples where: (1) pitch or roll exceeded ±5 °; (2)$pressurefelloutsidetherange[$
where $\overline{P}$ is the deployment median pressure; or (3) the minimum beam correlation dropped below 70%, following Nortek recommendations. Flagged samples were set to NaN.

A variability-based tilt quality control was applied rather than a fixed angular threshold. The rolling standard deviation of pitch and roll was computed over 60-second windows; samples where the variability exceeded 2 ° were flagged as unstable. An absolute tilt cap of 30 ° was retained for severely compromised instruments. Samples with stable but non-zero tilt (e.g., from pipes bent during storm events) were retained and corrected via a sample-by-sample three-dimensional rotation using the measured pitch (ϕ) and roll (θ) angles:

$$\mathbf{v}_{\text{corrected}} = \mathbf{R}_{\text{roll}}(\theta)\ \mathbf{R}_{\text{pitch}}(\phi)\ \mathbf{v}_{\text{raw}}, \tag{1}$$

where

$$\mathbf{R}_{\text{roll}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}, \qquad \mathbf{R}_{\text{pitch}} = \begin{bmatrix} \cos\phi & 0 & \sin\phi \\ 0 & 1 & 0 \\ -\sin\phi & 0 & \cos\phi \end{bmatrix}.$$

This tilt correction was applied before the heading rotation to buoy coordinates, ensuring that the velocity vector was first returned to the local vertical frame before projection onto the geographic coordinate system.

Velocity components were rotated from the instrument XYZ frame to a geographic buoy coordinate system ($+x$ west, $+y$ north, $+z$ up; left-handed, consistent with the MOP convention) using the magnetic heading from deployment notes and the magnetic declination computed via the International Geomagnetic Reference Field (IGRF-13) model at the deployment location and epoch.

1.3 Level 2: Spectral analysis

1.3.1 Segmentation and preprocessing

The continuous L1 time series were divided into non-overlapping 17.1 min segments of 2048 samples at 2 Hz. Segments in which more than 10% of pressure or velocity samples were flagged as NaN were discarded; for segments with ≤10% NaN, gaps were filled by linear interpolation before further processing. Prior to spectral analysis, velocities were rotated to a shore-normal coordinate frame ($+x$ onshore, $+y$ alongshore north) using the shore-normal angle retrieved from the CDIP THREDDS server for the corresponding MOP station.

For each valid segment, the mean pressure was used to compute the total water column depth $H = \overline{P} \times 10^4/(\rho g) + z_s$, where $\overline{P}$ is the segment-mean pressure in dBar, $\rho = 1025\ kg/m^3$,g $=$ 9.81 m/s$^2, and z_s$ is the pressure sensor height above the bed. Mean cross-shore, alongshore, and vertical velocities and temperature were recorded for each segment prior to detrending. The pressure time series was converted from dBar to meters of water head, and all channels (pressure, u, v, w) were linearly detrended.

1.3.2 Spectral estimation

Power spectral densities were estimated using Welch's method [welch1967] with 256-sample subsegments (128 s), a Hanning window, and 50% overlap, yielding approximately 34 effective degrees of freedom per segment. This configuration provides a frequency resolution of $\Delta f \approx 0.0078\ Hz$ over the range 0–1 Hz. Cross-spectra between pressure and the two horizontal velocity components were computed using the same windowing parameters for directional analysis.

1.3.3 Pressure correction

The measured pressure spectrum was converted to a surface elevation spectrum using the linear wave theory transfer function:

$$K_p(f) = \frac{\cosh(kz_s)}{\cosh(kH)} \tag{2}$$

where $k(f)$ is the wavenumber obtained by solving the linear dispersion relation $\omega^2 = gk\tanh(kH)$ via Newton's method. The surface elevation spectrum was computed as $S_\eta(f) = S_p(f)/K_p^2(f)$. Frequency bins where $K_p < 0.1$ were set to zero to suppress noise amplification at high frequencies; the resulting cutoff frequency f_{cut} was recorded for each segment.

1.3.4 Bulk wave parameters

Significant wave height was computed as $H_s = 4\sqrt{m_0}$, where $m_0 = \int S_\eta(f)\,df$ is the zeroth spectral moment, integrated separately over the sea–swell band (0.04–0.25 Hz) and infragravity band (0.004–0.04 Hz). Peak period $T_p = 1/f_p$ was determined from the frequency of maximum S_η within the sea–swell band, and the mean zero-crossing period was computed as $T_{m02} = \sqrt{m_0/m_2}$. The energy-weighted mean wave direction was obtained from the first-harmonic directional coefficients a_1 and b_1 derived from the pressure–velocity cross-spectra. Total energy flux was computed as $F = \rho g \int c_g(f)\,S_\eta(f)\,df$, where c_g is the group velocity from linear wave theory.

1.3.5 Near-bed velocity and stress

Near-bed orbital velocities were estimated by scaling the measured velocity spectrum at sensor height to the bed using linear wave theory. Each frequency bin of the velocity FFT was multiplied by $\cosh(kH)/\cosh[k(H-z_s)]$, and the result was inverse-transformed to obtain the near-bed velocity time series [[]e.g.,][]guza1980. The RMS near-bed orbital velocity U_b and wave orbital excursion A_w were computed from the bed velocity time series for each segment.

Wave-induced bed shear stress was estimated using the Swart (1974) friction factor formulation with a bed roughness of $k_s = 2.5D_{50}$, where $D_{50} = 0.25\ mm$ (to be updated with site-specific grain size distributions from laser particle analysis). Reynolds stress components ($\overline{u'w'}$, $\overline{v'w'}$, $\overline{u'v'}$)

and turbulent kinetic energy were computed from the detrended velocity fluctuations. Velocity moment statistics including skewness S_k and acceleration asymmetry A_s were computed following elgar1985,hoefel2003 for characterizing wave nonlinearity relevant to sediment transport.

1.4 Validation against MOP wave model

PUV-derived wave parameters were validated against hourly wave data from the CDIP Monitoring and Prediction (MOP) system, a spectral wave model initialized by offshore CDIP buoy observations. The MOP model provides wave energy density spectra $S_\eta(f)$ at 10 m depth along each MOP transect line. MOP spectra were shoaled to the PUV instrument depth using linear wave theory energy conservation:

$$S_\eta(f, h_{\mathrm{PUV}}) = S_\eta(f, 10\,\mathrm{m}) \times \frac{c_g(f, 10\,\mathrm{m})}{c_g(f, h_{\mathrm{PUV}})} \tag{3}$$

Shoaled MOP wave heights were computed by integrating the shoaled spectrum over the sea–swell band using the (non-uniform) MOP frequency bandwidth vector. PUV segment values (17 min resolution) were interpolated to the MOP hourly time grid for comparison. Agreement was quantified using R^2, RMSE, bias (PUV $-$ MOP), and the Willmott (1981) skill score.

Frequency-dependent validation was performed by computing the mean spectral ratio $S_\eta^{\mathrm{PUV}}(f)/S_\eta^{\mathrm{MOP,shoaled}}(f)$ across all matched hours, binned into low-swell (0.04–0.08 Hz), mid-swell (0.08–0.14 Hz), and high-swell (0.14–0.25 Hz) bands. This diagnostic isolates whether systematic biases originate from the pressure transfer function (which amplifies high-frequency energy most aggressively) or from other sources.

To investigate the origin of systematic biases, four candidate mechanisms were tested: (1) nonlinear shoaling via triad interactions, assessed by correlating the spectral ratio with the Ursell number $Ur = H_s L_p^2/h^3$; (2) directional narrowing during shoaling, assessed by comparing the first-harmonic directional spread $\sigma_1 = \sqrt{2(1 - \sqrt{a_1^2 + b_1^2})}$ between PUV and MOP; (3) bound long wave contamination, tested by examining the scaling of low-frequency excess energy with H_s^2 and the frequency structure of the excess relative to the expected group frequency; and (4) MOP model spectral peak broadening, quantified by comparing the Goda peakedness parameter $Q_p = (2/m_0^2) \int f\, S_\eta^2(f)\, df$ and the half-power spectral bandwidth between PUV and shoaled MOP spectra.

1.5 Level 3: Wave forcing characterization

1.5.1 Frequency-band energy decomposition

The L2 surface elevation spectrum was integrated over three frequency bands to characterize wave energy partitioning: infragravity (0.004–0.04 Hz), swell (0.04–0.10 Hz), and sea (0.10–0.25 Hz). Band-specific significant wave height, energy flux, and mean period were computed for each 17-minute segment. A dominant band flag was assigned based on which band carried the largest energy flux.

1.5.2 Storm event detection

Storm events were identified from the H_s time series using a threshold-exceedance method with configurable parameters (default: $H_s > 1.5$ m sustained for $\geq$6 hours, events separated by $<$12 hours merged). To characterize wave forcing during PUV data gaps (e.g., from instrument damage during storms), hourly MOP wave model data were loaded for a window extending 7 days beyond the PUV record. Storm events detected in the MOP record but absent from the PUV data were flagged as gap-period storms, providing continuous forcing context even when the in-situ instruments were compromised.

1.5.3 Transport proxies

Bottom energy flux was computed as the spectral integral of group velocity times surface elevation spectral density over the sea–swell band. The Shields parameter $\theta = \tau_b/[(\rho_s - \rho)gD_{50}]$ was computed per segment, with the critical Shields threshold from the Soulsby–Whitehouse formula. A mobilization flag indicated segments exceeding the critical threshold. Cumulative bottom energy flux was integrated over the deployment duration for comparison with morphological change between beach surveys. The Rouse number $P = w_s/(\kappa u_*)$ classified the dominant transport mode (bedload vs. suspended load).

1.5.4 Tidal current decomposition

Tidal harmonic analysis was performed on the segment-mean velocities using the `t_tide` package, with complex velocity input $(u + iv)$ enabling simultaneous analysis of cross-shore and alongshore tidal constituents. The tidal depth signal was obtained from NOAA tide predictions for the Scripps Pier gauge (station 9410230), which provided a smooth, gap-free tidal reference that outperformed the `t_tide` fit to the noisy PUV pressure record ($R = 0.995$ vs. $R = 0.987$ for TBR23). PUV instrument clocks were confirmed to be in UTC via cross-correlation with the NOAA record (optimal lag $= 0.0$ hours). Subtidal residual currents (observed minus tidal) captured wave-driven mean flows including undertow and longshore currents.

2 Results

2.1 PUV–MOP validation

2.1.1 Bulk wave parameter agreement

Of 40 instruments across 20 deployments, 33 were processed successfully (82.5%), with 7 failures attributable to hardware issues (burial, bent pipes, dead batteries). PUV-derived wave parameters showed strong agreement with the MOP wave model across all four TBR23 instruments (Table **??**). For instruments unaffected by biofouling, H_s correlations exceeded $R^2 = 0.83$ with RMSE $< 0.06\ m$ and biases of +0.002–+0.008 m. One instrument (MOP580 5m) exhibited lower agreement ($R^2 = 0.66$, bias = $+0.037\ m$), attributed to intermittent kelp fouling that reduced the valid segment rate to 49% compared with 85–91 for the remaining instruments. A small but consistent positive H_s bias was observed across all instruments, with PUV-measured wave heights systematically exceeding shoaled MOP values.

2.1.2 Frequency-dependent spectral bias

The frequency-resolved spectral ratio $S_\eta^{\mathrm{PUV}}(f)/S_\eta^{\mathrm{MOP,shoaled}}(f)$, averaged across all matched hours, revealed that the positive H_s bias was concentrated at the swell spectral peak (0.04–0.08 Hz), where the pressure transfer function $K_p > 0.9$ and the correction is minimal (Table **??**). The mid-swell band (0.08–0.14 Hz) showed near-unity ratios (0.99–1.03), and the high-swell band (0.14–0.25 Hz), where the pressure correction is most aggressive (K_p as low as 0.24), also showed only modest excess (1.03–1.05). The low-swell amplification increased with decreasing depth, from a band-averaged ratio of 1.17 at 8.3 m to 1.28 at 5.5 m. The overall excess energy relative to the shoaled MOP spectrum was 4.5 at 8.3 m depth and 8.7 at 5.5 m.

The concentration of excess energy at the spectral peak—where the pressure correction factor is close to unity—indicates that the bias does not originate from the pressure-to-surface-elevation transfer function.

2.1.3 Hypothesis testing

Four candidate mechanisms for the systematic positive bias were evaluated (Table **??**).

Nonlinear shoaling. The Ursell number $Ur = H_s L_p^2/h^3$, which characterizes wave nonlinearity and governs the strength of triad interactions during shoaling, showed no correlation with the low-swell spectral ratio ($R^2 = 0.004$, $N = 5228$ segment-hours across all instruments). The median Ursell number ranged from 12 at 8.3 m depth to 30 at 5.5 m, spanning conditions from weakly to moderately nonlinear.

Directional narrowing. The first-harmonic directional spread σ_1, computed from the PUV cross-spectral a_1 and b_1 coefficients, was systematically *wider* than the MOP model spread at swell frequencies (15 ° vs. 10 ° at the peak). This is opposite to the narrowing expected from refraction-driven focusing. No correlation was found between the spread ratio and H_s amplification ($R^2 < 0.001$). The wider PUV spread may reflect scattered wave energy at shallow depths not represented in the forward-propagating MOP model.

Bound long waves. Bound wave energy forced by wave group modulation scales as H_s^2 [longuethiggins1962]. The low-swell excess energy showed no positive correlation with H_s^2 ($R = -0.14$), and the amplification ratio weakly *decreased* with increasing wave height ($R = -0.19$). Furthermore, the excess energy was concentrated at the swell spectral peak (0.04–0.08 Hz), not at the expected group/bound wave frequencies (∼0.01–0.03 Hz).

MOP spectral peak broadening. The Goda peakedness parameter Q_p was systematically higher for PUV spectra than for shoaled MOP spectra (median $Q_p = 1.88$ vs. 1.75, ratio 1.07). The PUV half-power spectral bandwidth was 16.7 narrower than MOP (0.039 Hz vs. 0.047 Hz), and the PUV peak spectral density was 14.4 higher. The peakedness difference $\Delta Q_p = Q_p^{\mathrm{PUV}} - Q_p^{\mathrm{MOP}}$ showed the strongest correlation with H_s amplification of any tested mechanism ($R = 0.44$, $R^2 = 0.20$). Hours when the PUV spectrum was more peaked relative to MOP corresponded to hours with larger positive H_s bias.

These results indicate that the MOP spectral wave model introduces numerical broadening of the spectral peak during propagation from the initialization depth (10 m) to the instrument depth. The in-situ PUV measurement captures the true, narrower spectral peak shape, resulting in higher peak energy density and thus higher H_s for a given total variance. This spectral broadening is a recognized property of discretized spectral wave models [[]e.g.,][]rogers2002,booij1999 and has not been previously quantified using co-located PUV observations in this depth range.

2.1.4 Cross-deployment confirmation

The spectral peak broadening pattern was confirmed across 11 instruments spanning multiple deployments, sites, and seasons. The correlation between ΔQ_p and H_s amplification ranged from $R = 0.30$ to $R = 0.63$ across individual instruments, with all showing the same sign. These instruments spanned depths from 5.5 m to 15.5 m and included both summer and winter wave climates, indicating that the MOP model spectral broadening is a robust and general feature rather than a site-specific or seasonally dependent artifact.

2.1.5 Pipeline cross-validation: Ruby2D head-to-head

To verify that the multi-taper L2 implementation does not introduce biases relative to an established processing pipeline, we reprocessed the Ruby2D MOP582_6m record (Torrey Pines, October 2021 to February 2022, ∼3,100 hours) and compared it segment-by-segment against the archived L2 product produced by an independent prior pipeline (`beachpeeps/PUV_Processing`). The prior pipeline uses a single-periodogram estimator (Hanning window, full 60-minute segments, $\Delta f = 0.000278$ Hz) and applies a fixed cap of $K_p^2 \leq 10$ to the pressure transfer function. The legacy record has 3,095 60-min segments; the new pipeline produces 10,903 17-min segments over the same record. Segments were matched on midpoint nearest neighbor with a tolerance of 30 minutes, yielding 2,322 matched pairs (75%). The remaining 25% are hours that the new pipeline rejects under the segment-validity criterion (`nanMaxFrac` = 0.10) but that the prior pipeline retains by zero-padding small NaN gaps.

Bulk wave parameter agreement is excellent across all four metrics (Table 1; Figure 1). H_s agrees to 5 cm RMS with a −9 mm median bias ($R^2 = 0.98$), and direction agrees to within 1.2° RMS ($R^2 = 0.93$). The directional spread is 1° wider in the new pipeline on average; this is the largest systematic offset observed and remains within the segment-to-segment scatter. The new T_p shows the discrete quantization bands expected from $\Delta f = 0.000976$ Hz versus the prior pipeline's 0.000278 Hz, but no systematic bias.

Per-segment surface elevation spectra agree closely in shape (Figure 2). At a representative median segment ($H_s = 0.77$ m, November 27, 2021) and at a storm peak ($H_s = 2.78$ m, December 14, 2021, bimodal sea state), the integrated H_s values from the two pipelines agree to within

Table 1: Ruby2D MOP582_6m: bulk wave parameter agreement between the new multi-taper pipeline and the legacy archived L2. T_p statistics are computed after restricting both estimates to the SS band (4–25 s); without this filter, three segments on January 15, 2022 in the prior pipeline report T_p at the IG/DC bin ($T_p = 1200$, 1800, 3600 s) because the prior pipeline's peak picker is not band-limited.

Metric	Legacy (med.)	Multi-taper (med.)	Bias (M−L)	RMS	R^2
H_s (m)	0.771	0.757	−0.009	0.053	0.981
T_p (s)	13.48	13.30	−0.02	1.48	0.761
Direction SS (°)	−7.3	−7.2	+0.2	1.2	0.927
Spread SS (°)	14.7	15.8	+1.0	1.7	0.877

4%. The visible difference between the spectra is the chi-square noise characteristic of a single-periodogram estimator in the prior pipeline, versus the variance reduction provided by averaging $K = 7$ DPSS tapers in the new pipeline. The underlying peak shapes and locations agree.

This cross-validation establishes two things. First, the multi-taper implementation reproduces an independent prior pipeline to within instrument noise on bulk wave parameters, confirming that the estimator change is a refinement rather than a methodological break. Prior published Ruby2D bulk wave statistics do not require revision. Second, the consistency-of-windows fix in the new cross-spectral calculations (auto- and cross-spectra both use the same DPSS tapers, unlike the prior pipeline which mixed Hanning and Hamming) does not produce the order-20% directional errors that the bug analysis suggested were possible at this site — the practical impact at Ruby2D MOP582_6m is approximately 1° in spread.

2.1.6 Site-specific amplification at SIO Pier

The SIO Pier instruments (MOP 511, 7m depth) exhibited dramatically larger PUV-to-MOP spectral amplification than all other sites. The peak spectral density ratio ranged from 2.0 to 2.7 across four deployment periods (SIO24A through SIO25E), with corresponding H_s amplification of 29–36% — far exceeding the 1–11% observed at Torrey Pines, Solana Beach, and Los Peñasquitos sites at comparable depths.

Investigation confirmed that the MOP frequency grid is identical across all stations (20 bins, 0.04–0.40 Hz), ruling out a resolution artifact. The MOP model total energy at 10 m depth is comparable between SIO ($H_s = 0.62$ m) and Torrey Pines ($H_s = 0.61$ m), yet the in-situ PUV at SIO measures approximately 2.5× the peak spectral density predicted by the shoaled MOP spectrum.

The SIO Pier PUV is located near the head of Scripps Submarine Canyon, where canyon-edge refraction is known to create wave energy focusing [[]e.g.,][]munk1948,shepard1950. The canyon bifurcates nearshore, and wave focusing at the canyon head has been documented at Black's Beach on the adjacent fork. The MOP model, which propagates spectra along a relatively coarse along-shore grid, does not resolve this fine-scale bathymetric focusing. The PUV measurements capture the true, focused wave field, resulting in the observed amplification.

This site-specific effect is distinct from the universal spectral peak broadening described above. While broadening produces a modest (∼5–10%) Hs bias that is consistent across all sites, the canyon focusing at SIO produces an order-of-magnitude larger amplification that is unique to that location.

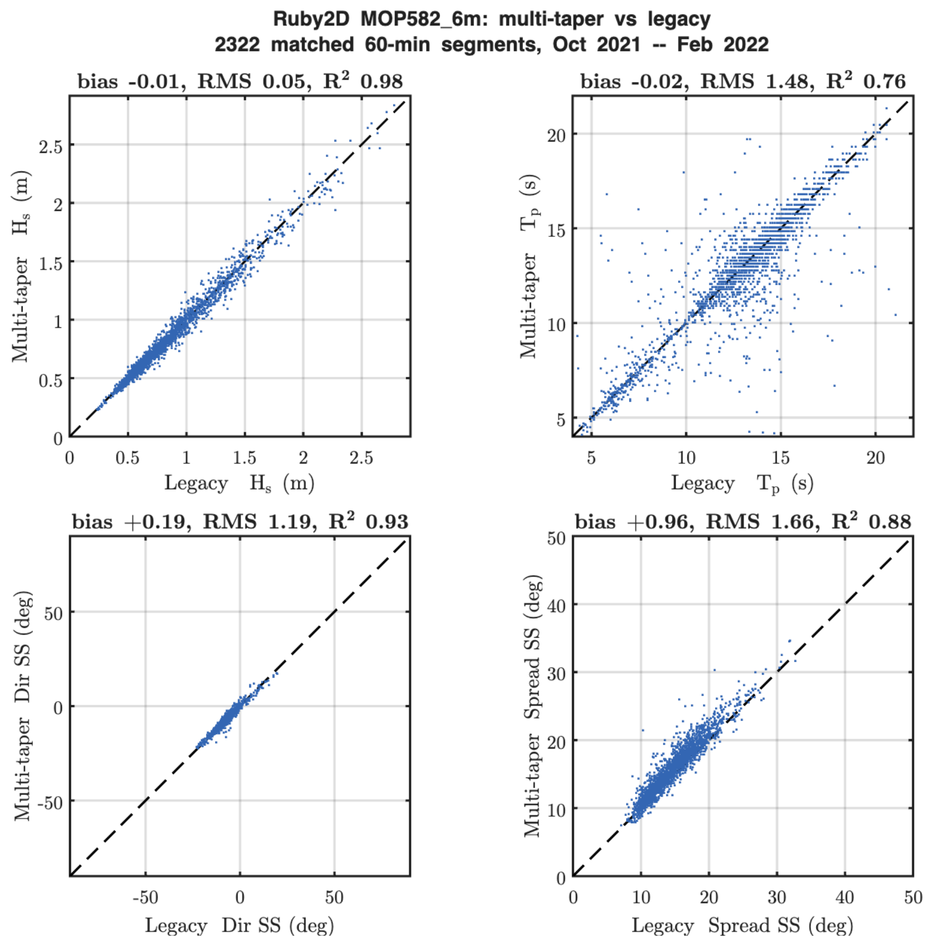

Figure 1: Bulk parameter agreement between the new multi-taper pipeline (vertical axis) and the legacy archived L2 (horizontal axis) for Ruby2D MOP582_6m, 2,322 matched 60-minute segments, October 2021 to February 2022. The dashed line is 1:1.

2.2 L3 wave forcing characterization

2.2.1 Energy band decomposition

Frequency-band analysis revealed clear seasonal patterns across all deployments. Winter deployments (TOR23W, SOL23, TOR24W, SOL24) were swell-dominated (65–90% of total energy flux in the 0.04–0.10 Hz band), while summer deployments (TBR23, SIO25B–D) showed approximately equal swell and sea contributions. The energy flux consistency check confirmed exact agreement between the sum of band fluxes and the total L2 energy flux ($F_{\text{total}}/F_{\text{L2}} = 1.0000$) for all 33 instruments.

2.2.2 Storm characterization with MOP gap-filling

Storm detection identified 0 events during the calm summer TBR23 deployment and up to 11 events during winter TOR23W, including the major December 28, 2023 storm sequence (H_s up to 3.6 m at 7 m depth, sustained over 74 hours). MOP gap-filling proved essential: the SOL23 MOP651 7m instrument detected 0 storms from PUV data (55% valid segments) but MOP identified 3 events

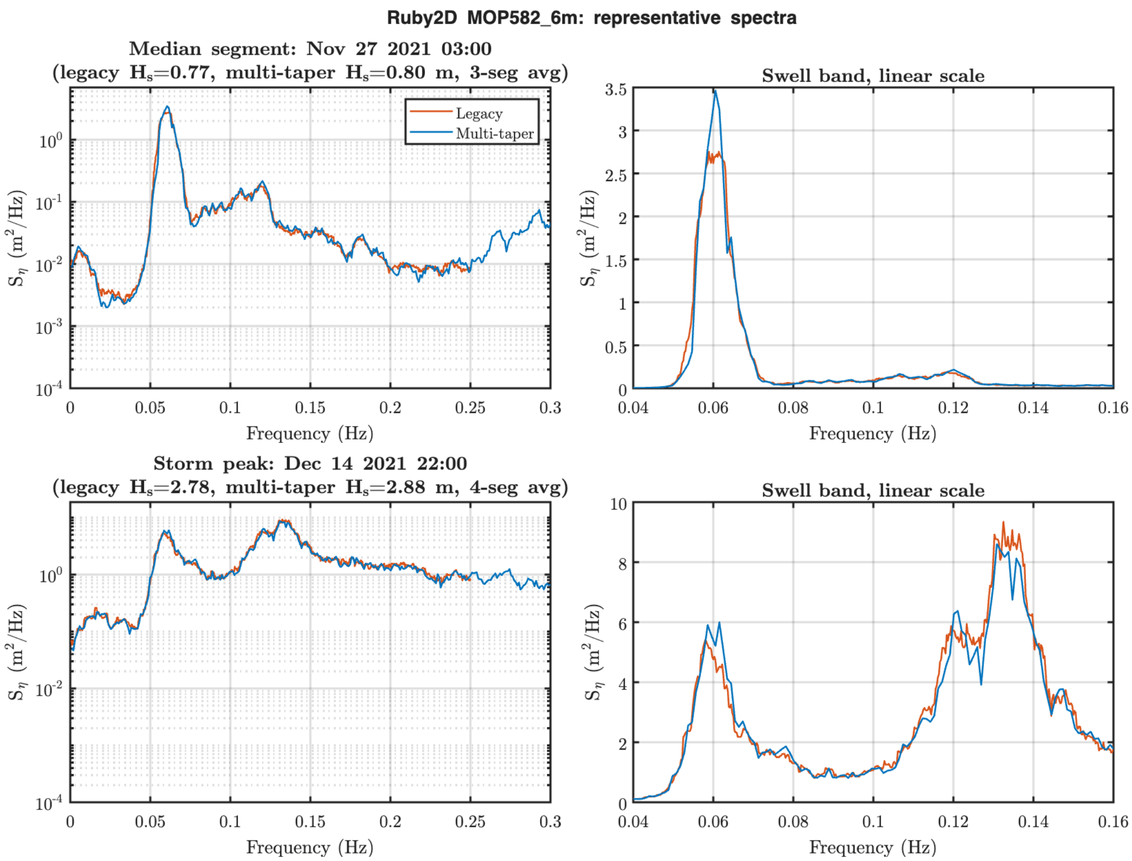

Figure 2: Representative surface elevation spectra for Ruby2D MOP582_6m. Top: median segment ($H_s = 0.77$ m). Bottom: storm peak ($H_s = 2.78$ m). Left column: log scale, full SS band. Right column: linear scale, swell band detail. The multi-taper estimates (blue) are smoother than the prior single-periodogram estimates (red); peak locations and total variance agree.

during data gaps, including an 86-hour storm with $H_s = 3.09$ m. Across all deployments, MOP supplemented 1–8 additional storm events per instrument that were missed due to PUV data gaps during the most energetic conditions.

2.2.3 Transport proxy patterns

The Shields parameter showed clear depth and seasonal dependence. Median Shields numbers decreased from 0.24 at 5.6 m to 0.065 at 15.5 m depth (TOR23W MOP586 cross-shore array), with corresponding mobilization fractions decreasing from 100% to 78%. Winter deployments showed approximately 3× the cumulative bottom energy flux of summer deployments at equivalent depths. The Rouse number indicated bedload-dominated transport (median $P = 2.8$–6.0) across all sites and seasons, with the lowest values (most suspension-capable) at the shallowest winter sites.

2.2.4 Tidal and subtidal currents

Tidal currents were small (typically <0.05 m/s) relative to wave-driven subtidal flows. The subtidal cross-shore residual was consistently negative (offshore-directed) across all sites and seasons, with magnitudes of 0.004–0.023 m/s, consistent with wave-driven undertow. Tidal depth predictions from the Scripps Pier NOAA gauge matched PUV depth anomalies at $R > 0.99$ for instruments with continuous records, confirming the UTC clock convention and validating the pressure-to-depth conversion.

2.3 TBR23 post-storm recovery dynamics

2.3.1 Observed morphological change

Jetski bathymetric surveys (22 surveys during the deployment period) revealed 25–39 cm of net erosion at the PUV instrument locations over the May–August 2023 deployment, with the 5 m sites eroding more than the 7 m sites. Pre-deployment context from the MOPS survey archive showed that the winter 2023 storm sequence had deposited approximately 85 cm of sand at the 5 m depth contour relative to the prior year, placing the bed well above its long-term equilibrium profile. The TBR23 deployment thus captured the post-storm readjustment phase rather than active storm erosion.

2.3.2 Competing transport mechanisms

Analysis of the PUV velocity moments revealed two competing cross-shore transport mechanisms operating simultaneously during the recovery period:

Onshore: velocity skewness. Velocity skewness S_k was positive at all three reliable instruments (median $S_k = +0.05$ to $+0.12$), indicating peaked crests and flat troughs characteristic of shoaling waves. Skewness increased with decreasing water depth, following the Ruessink et al. (2012) parameterization based on the Ursell number. The skewness-driven bedload flux $\langle u|u|^2 \rangle$ was directed onshore, representing the wave-driven recovery mechanism.

Offshore: undertow. The subtidal mean cross-shore current was consistently offshore-directed ($\overline{u} = -0.006$ to -0.012 m/s) at all three instruments, representing wave-driven return flow (undertow). Undertow magnitude increased with wave height and was modulated by tidal phase, with stronger offshore flow at high tide when more wave energy reached the instrument depths.

2.3.3 Depth-dependent transport balance

The ratio of onshore skewness flux to offshore undertow flux showed a clear depth dependence exploiting the tidal modulation of water depth at each fixed instrument location ($\sim$2.5 m tidal range). At the shallowest site (MOP586 5m, median depth 5.5 m), the skewness flux exceeded the undertow flux by a factor of 3–4, indicating net onshore transport and active recovery. At the 7 m sites (median depth 8.3 m), the undertow flux slightly exceeded the skewness flux, indicating near-equilibrium or weak offshore transport.

The crossover depth where onshore and offshore fluxes balanced was approximately 6–7 m, defining a *recovery boundary* below which the post-storm deposit was actively being redistributed shoreward. This is consistent with the survey observations: the MOP586 5m site exhibited a 25 cm recovery pulse in mid-July 2023 that was not observed at the deeper instruments.

One instrument (MOP580 5m) was excluded from the transport analysis due to intermittent kelp fouling that produced anomalous negative velocity skewness (92% of segments) inconsistent with the other instruments and with wave theory.